Virtual Temperature Monitoring PCB

Design & Verification Report

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1. Introduction

This report presents the engineering design and computer-based verification of a virtual temperature-monitoring system. The project covers the main stages of the hardware development process, including schematic capture, analog signal conditioning, ADC analysis, power regulation, microcontroller interfacing, SPICE simulation, and PCB layout. The aim is to develop and document a complete embedded temperature-monitoring system using KiCad 10 and computer-based simulation tools.

The project is limited to computer-based design and verification. No physical PCB was fabricated, assembled, or experimentally tested. The results therefore represent a virtual prototype rather than measured hardware performance.

2. System Architecture

The system consists of a virtual temperature sensor, an RC filter, a microcontroller for signal processing, and a regulated power supply. The virtual sensor produces an analog voltage proportional to the modeled temperature. This signal passes through the RC low-pass filter before being connected to the STM32G431CBUx ADC through the ADC_TEMP input. The MCU handles the digital processing and provides a UART interface to a conceptual host PC.

The main signal path is: Temperature → Virtual Sensor → RC Filter → ADC_TEMP → STM32 → UART → PC

The power subsystem takes a nominal 5 V input and regulates it to approximately 3.3 V using an AMS1117-3.3 linear regulator. The 3.3 V rail supplies the STM32 MCU and the supporting circuitry, including the local decoupling capacitors.

The main power path is: 5 V Input → AMS1117-3.3 → 3.3 V Rail → STM32 and Supporting Circuitry.

3. Design Requirements

Parameter	Specification / Component	Details
Temperature Range	0 to 100 °C	Target operational range
Sensor Voltage Range	0 to 2.0 V	Linear analog output
ADC Reference / Supply	3.3 V	Provides 1.3 V headroom
ADC Resolution	12-bit (4096 levels)	Assumed internal MCU ADC
Microcontroller	STM32G431CBUx	QFN-48 package
Power Supply	5.0 V Input → 3.3 V Output	AMS1117-3.3 LDO regulator
Filter Components	R1 = 1 kΩ, C1 = 100 nF	Low-pass filter
Interface	UART (TX, RX, GND)	External communication

4. Temperature Sensor Model

For the simulation, the temperature sensor is represented using a simple linear relationship between temperature and output voltage:

$$V_{TEMP} = 0.02 \times T$$

Where V_{TEMP} is the output voltage in Volts and T is temperature in °C. This gives a sensor sensitivity of 0.02 V/°C, so the modeled sensor produces 0 V at 0 °C and 2.0 V at 100 °C.

Temperature (T)	Sensor Output (V_TEMP)
0 °C	0.0 V
25 °C	0.5 V

Temperature (T)	Sensor Output (V_TEMP)
50 °C	1.0 V
75 °C	1.5 V
100 °C	2.0 V

5. Analog Signal Conditioning

To reduce high-frequency noise and smooth rapid voltage changes before digitisation, a passive single-pole RC low-pass filter is used. The filter consists of R1 = 1 kΩ and C1 = 100 nF and produces the filtered signal, TEMP_FILTERED. The main filter characteristics are:

- Time Constant (τ): $\tau = R \times C = 1000\,\Omega \times 10^{-9} F = 100\,\mu s$
- Cutoff Frequency (f_c): $f_c = \frac{1}{(2\pi)(R \times C)} = 1.59\,kHz$

The filter smooths voltage transitions while still allowing the relatively slow changes expected from a temperature sensor to pass through.

For a step change in sensor voltage, the theoretical filtered response is:

$$V_{FILTERED}(t) = V_{TEMP}\left(1 - e^{\frac{-t}{RC}}\right)$$

The output therefore approaches the new sensor voltage gradually rather than changing instantaneously. This helps reduce rapid voltage changes and high-frequency noise before the signal is measured by the ADC.

6. Power Supply Design

A 3.3 V power rail is used to supply the digital and analog parts of the circuit.

- Input Regulation: Nominal 5V input via connector J2 regulated to 3.3V by an AMS1117-3.3 LDO (U2).
- Filter Capacitors: 10 μF input capacitor (C2) and 10 μF output capacitor (C3) placed at the input and output of the regulator.
- MCU Decoupling: Three 100 nF ceramic capacitors (C4, C5 and C6) are placed near the

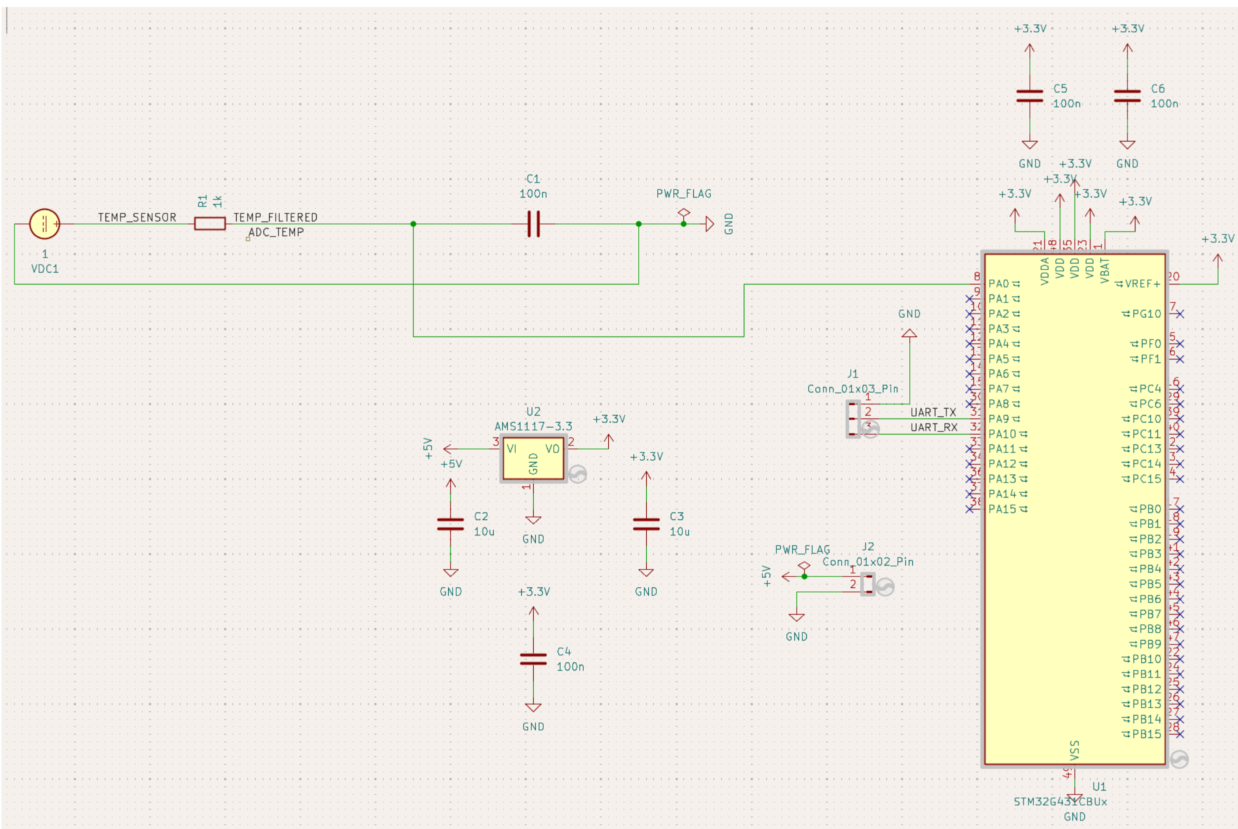
STM32 power pins to help reduce supply noise caused by switching.

7. Microcontroller & Schematic Capture

The schematic uses an STM32G431CBUx microcontroller (U1) to handle the analog measurement and digital processing.

- Pin Mapping: The filtered temperature signal is connected to the ADC input, while UART communication is connected to the UART TX and RX lines on header J1.

Figure 1. Complete KiCad Schematic



The complete KiCad schematic showing power distribution, MCU connections, filtering, and UART header.

Electrical Rules Check (ERC) Results

The completed schematic was checked using KiCad's Electrical Rules Check (ERC). The final schematic produced 0 ERC errors, confirming that the electrical connections and power-related connections met the project's schematic rules.

Final ERC result: 0 errors

8. Firmware Implementation and Verification

The STM32G431CBUx firmware was developed in STM32CubeIDE using the STM32 HAL. ADC1 is set up to read the filtered temperature signal on PA0 (ADC1_IN1) using a 12-bit, single-ended conversion. USART1 is configured on PA9 (TX) and PA10 (RX) with a baud rate of 115200 bit/s for serial output.

ADC-to-Temperature Conversion

The firmware converts the measured ADC code into an input voltage using the 3.3 V ADC reference:

$$V_{ADC} = \left(\frac{ADC_{CODE}}{4095}\right) \times 3.3$$

The temperature sensor model used in the design is:

$$V_{TEMP} = 0.02 \times T$$

Therefore, the firmware calculates temperature using:

$$T = \frac{V_{TEMP}}{0.02}$$

The intended temperature range is 0–100 °C, which corresponds to a sensor output of 0–2.0 V. Since the ADC can measure up to approximately 3.3 V, the same sensor model would give a maximum theoretical temperature of:

$$T_{max} = \frac{3.3V}{0.02} = 165°C$$

The firmware therefore allows temperature conversion over the full theoretical ADC range of 0 - 165 °C while the intended system operating range remains 0 - 100 °C.

UART Reporting

After each ADC measurement, the firmware calculates the voltage and temperature and sends the results through USART1. The output includes the raw ADC value, voltage, and calculated temperature. For example:

ADC: 1241, Voltage: 1.000 V, Temperature: 50.00 C

Firmware Verification

The firmware was successfully built in STM32CubeIDE for the STM32G431CBUx. The final Debug build completed with 0 errors and 0 warnings and produced ELF, HEX, and binary files.

The firmware has not yet been tested on the actual PCB because the board has not been fabricated and assembled. Therefore, the ADC readings, temperature calculation, and UART output still need to be checked on physical hardware.

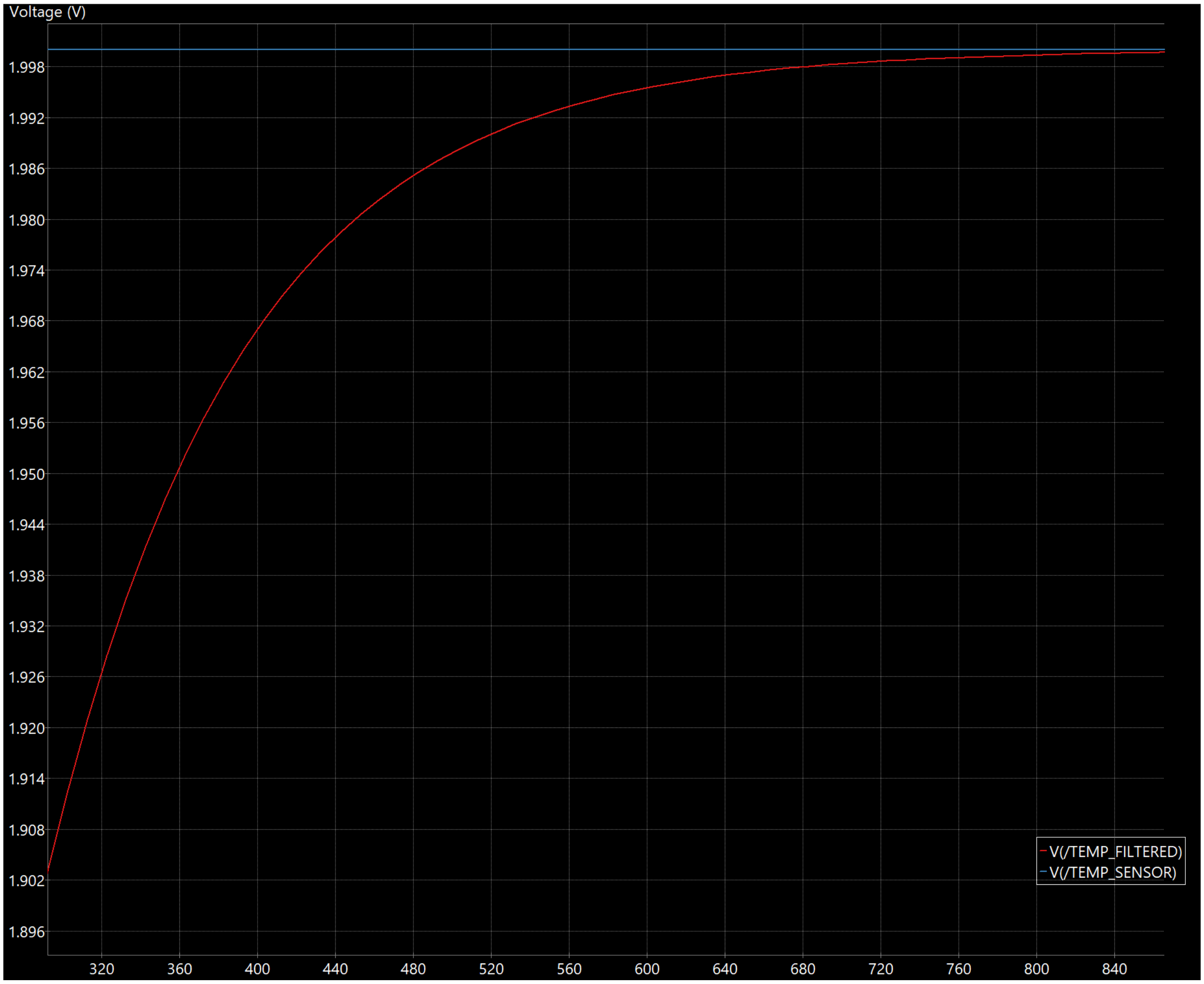
9. SPICE Simulation and Results

Transient SPICE simulations were run to verify time-domain signal conditioning through the RC filter at representative operating voltages.

Figure 3. Filter Transient Response at 1.0 V Input (50 °C equivalent)

The simulation demonstrates the expected low-pass behavior, with the filtered signal *transitioning smoothly toward the applied 1.0 V sensor voltage over several time constants.*

Figure 4. Filter Transient Response at 2.0 V Input (100 °C equivalent)



At maximum operating temperature, the filter settles smoothly at 2.0 V, well below the 3.3 V ceiling, proving signal integrity without clipping.

10. ADC Resolution & Quantization Analysis

With a 12-bit ADC (2^{12} = 4096 levels) and a 3.3V reference voltage (V_REF), the voltage resolution per LSB is:

$$LSB\ Step\ =\ \frac{3.3\ V}{4096} = 0.806\ \frac{mV}{count}$$

The expected output code is determined by: $ADC_{Code} = \left(\frac{V_{ADC}}{3.3}\right) \times 4095.$

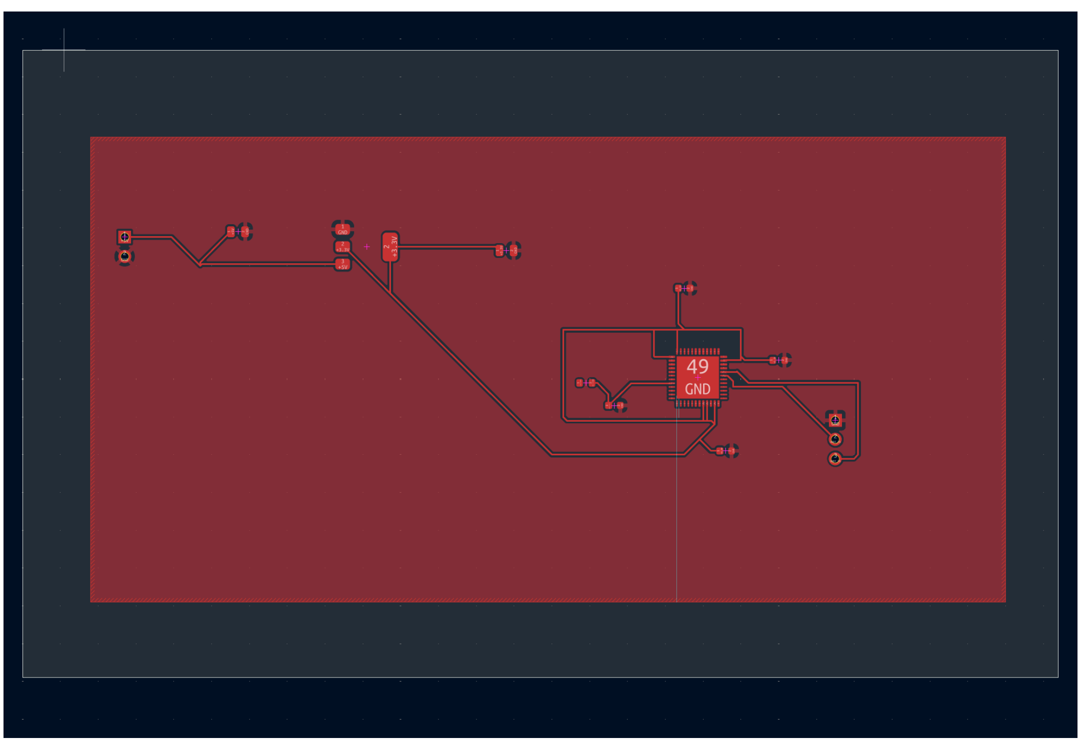
Temperature (T)	Voltage (V_ADC)	ADC Calculation	Expected Code
0 °C	0.0 V	(0.0 / 3.3) × 4095	0
25 °C	0.5 V	(0.5 / 3.3) × 4095	620
50 °C	1.0 V	(1.0 / 3.3) × 4095	1241
75 °C	1.5 V	(1.5 / 3.3) × 4095	1861
100 °C	2.0 V	(2.0 / 3.3) × 4095	2482

11. PCB Design & Layout

Components were arranged according to functional signal flow: Power Entry → Regulation → Sensing & Filtering → MCU → Output Header.

- Placement & Footprints: 0603 passive components (R1, C1, C4 - C6), 0805 power caps (C2, C3), SOT-223 regulator (U2), and QFN-48 MCU (U1).
- Grounding: Copper zones assigned to GND provide low-impedance return paths.

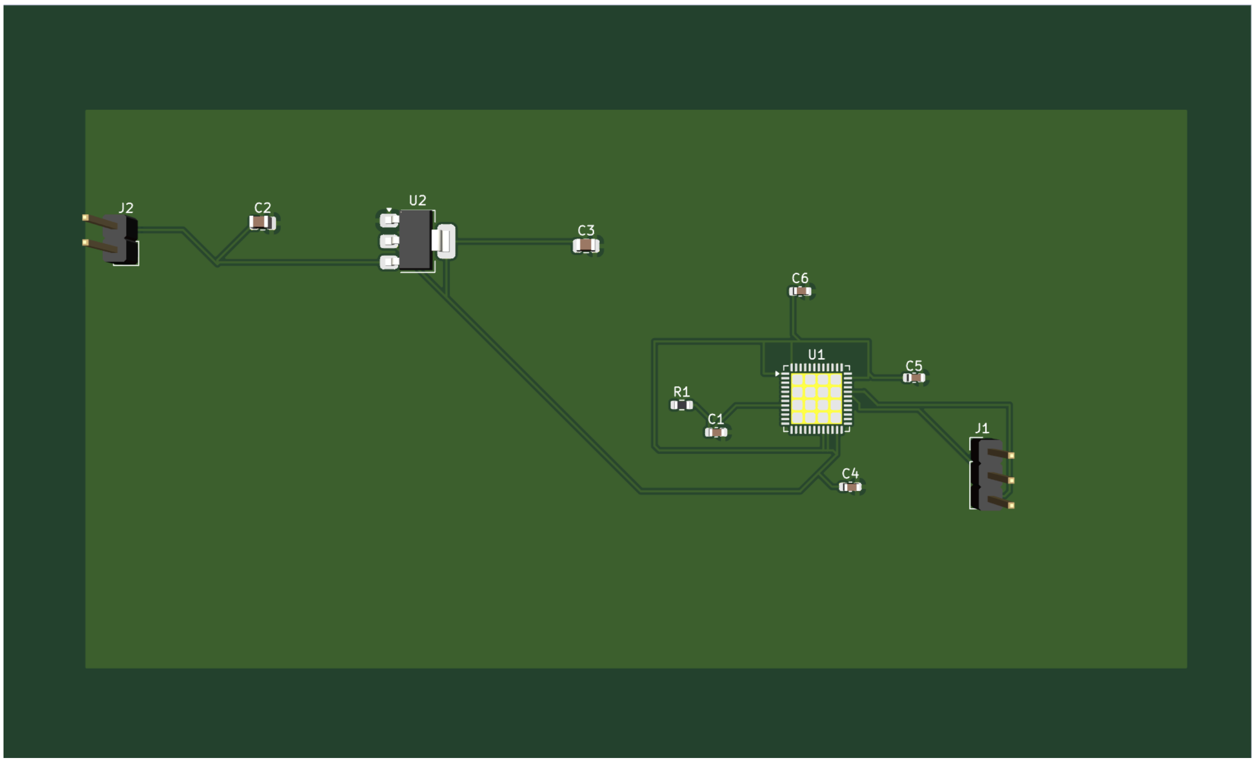
Figure 5. KiCad PCB Layout



Layout routing details showing power rails, signal paths, and copper ground plane fill.

12. Verification & Manufacturing Outputs

Figure 6. 3D PCB Model Visualization



3D model render displaying component placement, footprint alignments, and board geometry.

Bill of Materials (BOM)

RefDes	Description / Part	Footprint / Value
R1	Signal Resistor	0603 / 1 kΩ
C1, C4, C5, C6	Ceramic Capacitors	0603 / 100 nF
C2, C3	Tantalum/Ceramic	0805 / 10 µF

RefDes	Description / Part	Footprint / Value
	Capacitors	
U1	Microcontroller	QFN-48 / STM32G431CBUx
U2	LDO Linear Regulator	SOT-223 / AMS1117-3.3
J1, J2	Terminal Connectors	Header Pin / 1x3 (UART), 1x2 (Power)

13. Design Limitations

The temperature sensor was represented using the assumed mathematical model $V_{TEMP} = 0.02 \times T$, rather than a characterized physical sensor. Therefore, actual sensor accuracy, offset, linearity, and temperature drift were not evaluated.

STM32 firmware was implemented using STM32CubeIDE and successfully compiled for the STM32G431CBUx microcontroller. The firmware configures ADC1 for measurement of the filtered temperature-sensor signal and USART1 for serial reporting. The final build completed with 0 errors and 0 warnings and generated ELF, HEX, and binary firmware images. However, hardware-level verification of ADC measurements, temperature conversion, and UART communication remains pending because the physical PCB has not yet been fabricated and assembled.

A remaining GND-zone connectivity issue was identified during PCB DRC and was not fully resolved. The project is therefore considered a virtual PCB design and simulation study rather than a fully manufacturing-verified hardware design.

14. Lessons Learned

This project provided experience with the complete PCB design workflow, from system architecture and schematic capture through simulation, footprint assignment, PCB placement, routing, verification, 3D visualization, and manufacturing-file generation.

Key concepts practiced included RC filtering, ADC analysis, power regulation, MCU decoupling, grounding, PCB routing, ERC/DRC verification, and manufacturing documentation. The project also provided experience with STM32 firmware development, including ADC configuration, temperature conversion, and UART communication.

15. Conclusion

A complete virtual temperature-monitoring PCB was designed using KiCad. The system models a 0–100 °C temperature range using a 0–2 V sensor signal, which is filtered through a 1 kΩ / 100 nF RC network before being measured by the assumed 3.3 V, 12-bit ADC of the STM32G431CBUx.

SPICE simulation showed the expected RC transient response, with the filtered signal approaching the applied sensor voltage. The schematic, PCB layout, power system, UART interface, footprints, ground planes, 3D model, manufacturing files, BOM, and STM32 firmware were completed. The firmware also compiled successfully with 0 errors and 0 warnings.

Since the project was completed entirely in software, the design has not yet been tested on physical hardware. Before fabrication, the remaining GND-zone DRC issue should be resolved. The PCB can then be fabricated and assembled, after which the firmware and hardware can be tested to verify ADC measurements, temperature conversion, UART communication, and operation over the intended 0–100 °C range.

Appendix A — Key Equations

- Sensor model: $V_{TEMP} = 0.02 \times T$
- Temperature from sensor voltage: $T = \frac{V_{TEMP}}{0.02}$
- RC time constant: $\tau = R \times C = 1000\,\Omega \times 10^{-9} F = 100\,\mu s$
- Cutoff Frequency f_c: $f_c = \frac{1}{(2\pi)(R \times C)} = 1.59\,kHz$
- Filtered Voltage: $V_{FILTERED}(t) = V_{TEMP}\left(1 - e^{\frac{-t}{RC}}\right)$
- ADC number of levels: $2^{12} = 4096$
- ADC resolution: $\frac{3.3\,V}{4096} = 0.806\,\frac{mV}{count}$
- ADC conversion: $ADC_{Code} = \left(\frac{V_{ADC}}{3.3}\right) \times 4095$